

AST326 LAB REPORT 5: MAPPING THE MILKY WAY

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ABSTRACT

We analyze the galactic rotational curve by using the 21 cm neutral hydrogen at various galactic longitudes. We find the peak velocities by using the Doppler equation and the rotational velocity using the Tangent point method. When plotting our results we find a linear relationship curve with a decreasing change in rotational velocity as the galactocentric radius increases.

1. Introduction

The Milky way is a spiral galaxy with a rotating disk containing stars, gas, planets, and dust. To understand the structure and dynamics our galaxy, it is useful to map the distributions and motion of gas. This understanding is crucial to study the galaxies mass distribution, especially due to various parts of the galaxy not following a standard Keplerian system. We expect rotation velocity to decrease with distance from the Galactic center. However we find that the rotation curve is flat which is most likely due to the presence of Dark matter. (Sofue & Rubin 2001) Due to H being the most abundant element present in the universe, we will use its emission line, the 21cm emission line, to find the Doppler shift which will then in turn give us the radial velocities.

The Doppler shift, or Doppler effect is a formula that describes the change in frequency observed or emitted by an object due to relative motion. When the object moves towards the observer, it experiences a blueshift, while if it moves away, it experiences a redshift. This motion can be describe by the formula below:

$$\frac{v_r}{c} = \frac{f_{\text{obs}} - f_0}{f_0} \quad (1)$$

where v_r is the radial velocity of the source relative to the observer, and c is the speed of light.

In this report, we will use the 21 cm lab to find the blue and red shift of the hydrogen line at various longitudes. The Data was collected using the MP horn at the university of Toronto. It was equipped with an antenna and an AirSpy 2G receiver to reformat and digitize the data taken in into .dat files. We will then use median filtering to remove the baseline and take dark images to, at least partially, remove signals created by the environment around us. Finally, we will find the radial velocities and use them along the the tangen point method to find the Galactic rotation curve between 0 and 8 kpc.

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2. Observation and Data

As stated in the introduction, we have decided to use the MP Horn telescope situated at the University of Toronto. It was equipped with 0.5m horn antenna. To take in the data we attached an AirSpy 2G spectrometer that was then linked with a computer.

To record in the observations, we decided to pick a frequency of 1420MHz, which will be very close to the 21 cm hydrogen line. The range we will capture over is from about 1418 MHz to 1422 MHz. Although it may be extreme, we decided to record over a wide range of frequencies: Starting at a galactic longitude of 10° , we increment the longitude by 3° until we reach 90° (in our case, we read 88°). We started taking the data at 10:35 AM EST. To see full table of date and times with galactic coordinates, see appendix section 1.

We then performed a secondary measurement to measure the darks at different angles, starting at a longitude of 28° , and ending at 82° by incrementing by 9° . We then do an extra measurement at our max, so at 88° .

The data is given in a .dat file type with the format of:

- 4 Bytes $\rightarrow$ INT32, nf = number of frequency channels
- 4 Bytes $\rightarrow$ INT32, nt = number of time samples
- 4 Bytes $\rightarrow$ FLOAT32, alt = telescope pointing at time of save
- 4 Bytes $\rightarrow$ FLOAT32, az = telescope pointing at time of save
- 4 Bytes x nf $\rightarrow$ FLOAT32, lf = list of frequencies
- 8 Bytes x nt $\rightarrow$ FLOAT64, lt = list of unixtimes
- 4 Bytes x nf $\rightarrow$ FLOAT32, sp = averaged spectrum
- 4 Bytes x nf $\rightarrow$ FLOAT32, bl = spectral baseline
- 4 Bytes x nf x nt $\rightarrow$ FLOAT32, wf = waterfall samples

To make sure it was taken correctly, we use the waterfall plot data using the waterfall samples and the list of frequencies.

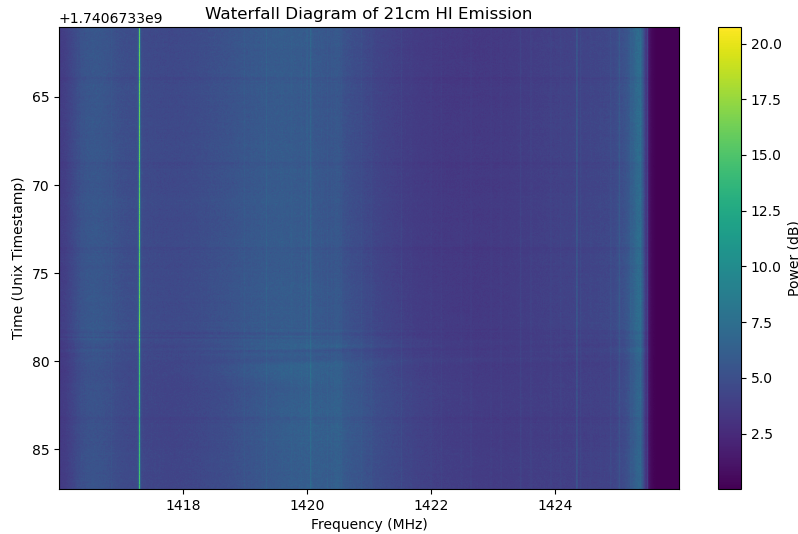


Fig. 1.—: Waterfall plot of the observed spectra. We chose the 26th data set to use as an example

3. Data Methods and Reduction

3.1. Experimental Method

3.1.1. Baseline removal

After loading the dictionaries, we organized the 27 data sets into a list and stored the 6 dark frames separately. To visualize the raw spectral data, we plotted a frequency diagram using the `lf` (list of frequencies) and `sp` (averaged spectrum). This provided an initial graph of the raw spectrum.

Upon inspection, we noticed that the spectrum was not baseline-corrected (i.e., it did not fluctuate around zero). To address this, we needed to determine and subtract a baseline.

Several methods exist for baseline correction, with polynomial fitting (`polyfit`) from the SciPy library being a common approach. However, when attempting to fit the entire curve, we found that polynomial fitting was sensitive to random signal variations, leading to an inaccurate baseline. While masking out strong signals improved the fit, there were too many interfering signals to manually mask effectively.

To mitigate this issue, we opted for a different SciPy function: median filtering. This method divides the data into subsets of length n , sorts each subset, and replaces each point with the median of its respective subset. This approach is particularly useful for large datasets as it reduces the influence of outliers while preserving the overall trend. We found that using a window size of $n = 50$ produced a significantly improved baseline fit compared to polynomial fitting.

3.1.2. Radial velocity

To calculate the radial velocity, we identify a small peak in the spectrum within the frequency range of 1420 to 1421 MHz. The frequency is then converted into velocity using the Doppler shift equation:

$$\frac{f_0 - f}{f} = \frac{v_r}{c} \quad (2)$$

Throughout this process, we observed that the detected peak consistently falls within the velocity range of -1 to -28 km/s for our dataset. To systematically identify these peaks, we use the `find_peaks` function from the SciPy library, selecting the four most prominent peaks within this region.

3.1.3. Galactic rotation

To find the galactic rotation, we will use the Tangent point method, where we first choose which peak we will use among the peaks found with Scipy. We have decided to use the maximum velocity, so using the rightmost peak, or the one closest to the -28 axhline (axhline is a Matplotlib function). We can then use this v .

$$v_{\text{rot}} = v + v_{\text{sun}} \sin(l)$$

$$d = R_0 \sin(l)$$

where:

- v_{rot} is the rotational velocity.
- v is the velocity of the object in the direction of galactic rotation.
- v_{sun} is the Sun's rotational velocity.
- l is the Galactic longitude.
- d is the distance of the object from the Sun.
- R_0 is the Sun's distance from the Galactic Center.

We can then plot a d by v_{rot} graph which will show how an object will rotate around the milky way.

3.2. Data Reduction

In this section we will go over the various ways we have reduced our data set to find the data we need.

The MP horn is a telescope at the University of Toronto and is surrounded by a large amount of signals, creating a large amount of noise that we do not want. We first reduce the data set by taking the average of all our dark samples and making them into one file. We then iterate through every data set and remove this mean dark.

In addition, the Airspy has a signal that it transfers in the data set at exactly 1420 MHz. Although it is partially removed by the darks, we still see a large signal at this value, so these values will be masked. Masking data is the action of not taking into account a data point, usually an outlier, while replacing it to a more sensible value.

The baseline subtraction uses median filtering, which removes the curve and sets the frequencies to zero. We can then see the lump and limit both the x and the y axes to only see this section.

The final reduction occurs when we choose the peaks. In this report we have decided to choose the peak with the maximum velocity, or the rightmost peak.

4. Analysis and Results

As it is hard it see what we are doing with many spectrum on the diagrams, we will choose a data set and use it as an example. For 4.3 and the end for 4.2, we will iterate over all the angles and redo the same method for results. In this example, we will choose the 26th data set.

4.1. Frequencies and baseline subtraction

will start by opening the file and plotting the raw data.

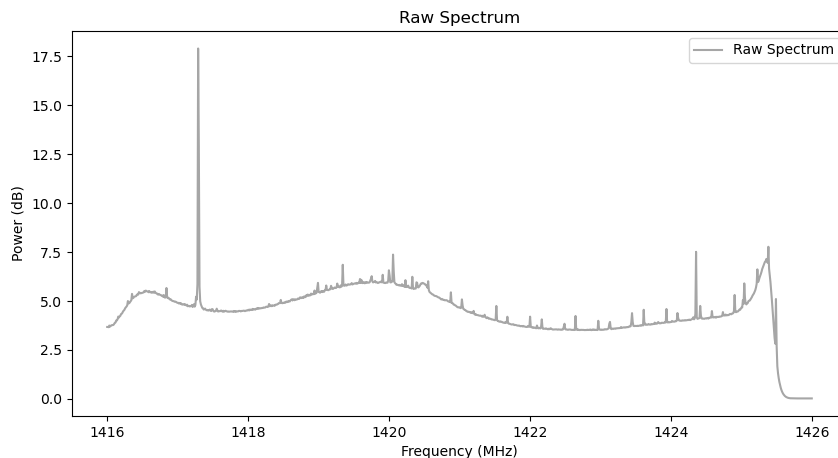


Fig. 2.—: Raw spectrum of 26th data set

We then remove the average of the Darks and trace a baseline onto the spectrum.

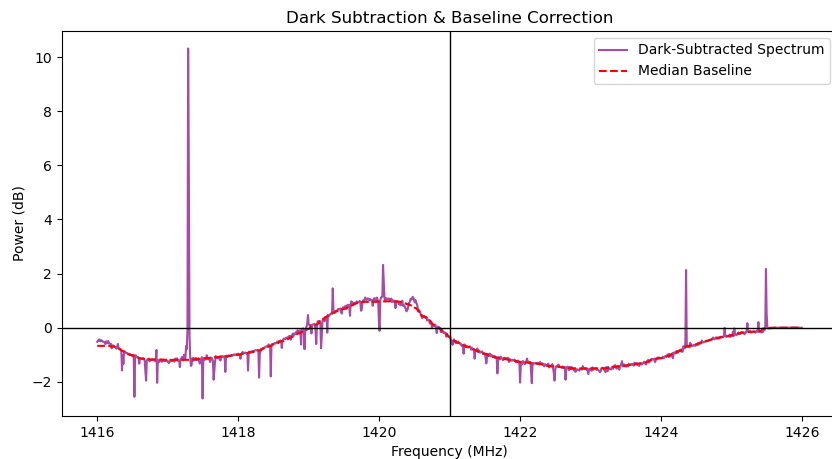


Fig. 3.—: Baseline fitting and Dark subtraction of 26th data set

We can now see the new corrected spectrum that will be used for analysis of radial velocity. The vertical line in both diagrams is the 1421 MHz line

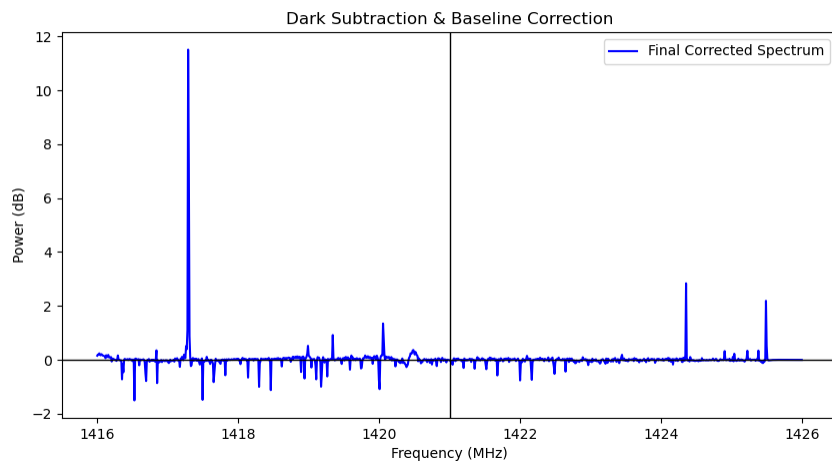


Fig. 4.—: Corrected spectrum of 26th data set

4.2. Radial Velocities

From Figure 5, we see that there is a lump to the left of the 21 cm line. We zoom in on the lump after performing the velocity calculation.

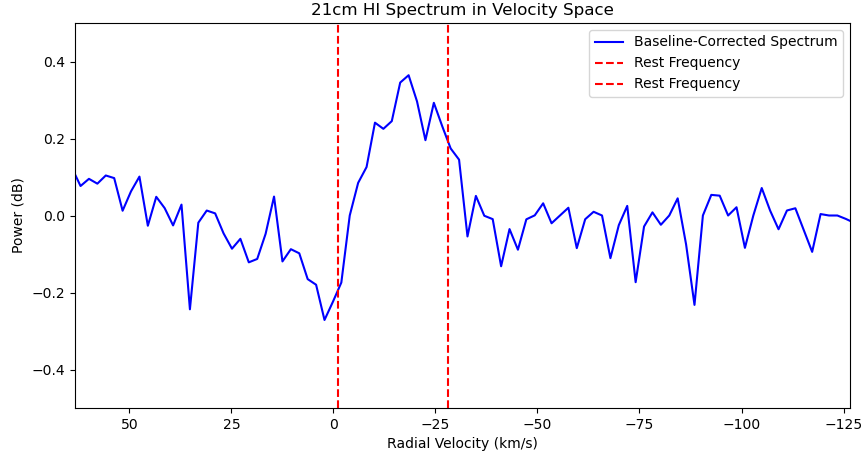


Fig. 5.—: Radial velocity calculations of 26th data set

We put in place 2 red dotted lines that limit the lump. These are consistent throughout all our data sets, where the lump is between -1 and -28 km/s. We can then use the find peaks function in this range.

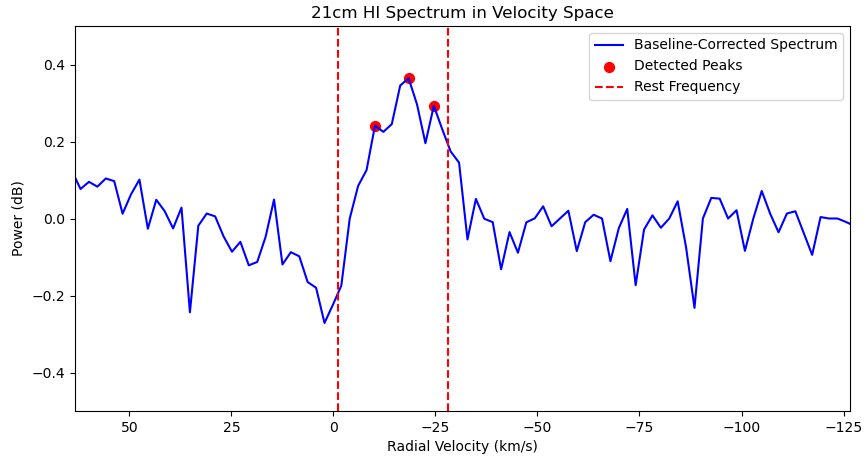


Fig. 6.—: Finding and Marking peaks of velocity in 26th data set

In this case, we see that there are only 3 peaks. As we choose the rightmost one, as long as there is at least one peak, the data set is valid.

4.3. Galactic coordinates

After performing the point tangent method to find the R and V values for the galaxy, we will tabulate these along with their corresponding frequencies and radial velocities

R_values	V_values	Peak Velocities	Peak Frequencies
1.420095	12.749672	-25.452928	1420.526367
1.839650	28.157943	-21.331289	1420.506836
2.254162	37.248096	-23.392123	1420.516602
2.662496	46.172066	-25.452928	1420.526367
3.063533	59.021328	-23.392123	1420.516602
3.456172	69.583895	-23.392123	1420.516602
3.839338	81.952455	-21.331289	1420.506836
4.211981	85.794672	-27.513704	1420.536133
4.573080	95.508735	-27.513704	1420.536133
4.921643	109.007183	-23.392123	1420.516602
5.256717	118.021152	-23.392123	1420.516602
5.577383	122.525935	-27.513704	1420.536133
5.882761	130.741052	-27.513704	1420.536133
6.172015	142.643985	-23.392123	1420.516602
6.444352	149.970243	-23.392123	1420.516602
6.699025	156.821327	-23.392123	1420.516602
6.935337	163.178459	-23.392123	1420.516602
7.152640	169.024213	-23.392123	1420.516602
7.350338	174.342568	-23.392123	1420.516602
7.527889	174.997364	-27.513704	1420.536133
7.684806	179.218672	-27.513704	1420.536133
7.820660	186.994924	-23.392123	1420.516602
7.935078	192.133771	-21.331289	1420.506836
8.027747	192.565858	-23.392123	1420.516602
8.098412	194.466853	-23.392123	1420.516602
8.146880	195.770711	-23.392123	1420.516602
8.173018	196.473859	-23.392123	1420.516602

Table 1:: Table of R values, V values, peak velocities, and peak frequencies for all data sets

We then plot and fit a Rotation velocity (V) vs Galactocentric Radius (R) graph.

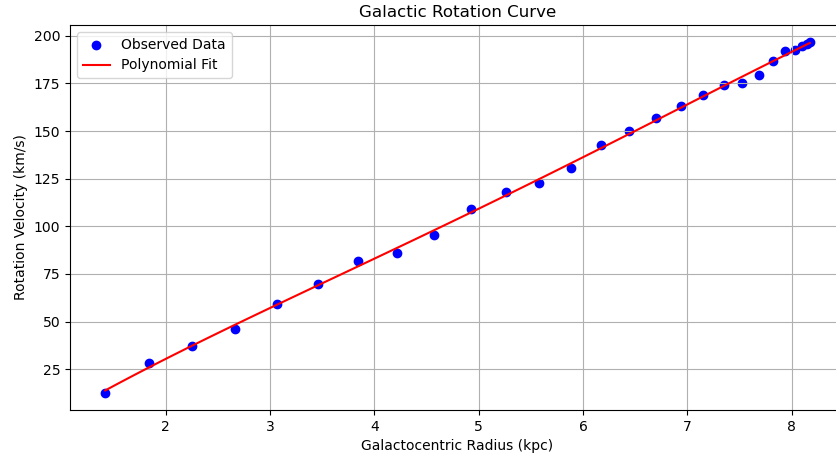


Fig. 7.—: Calculated and fitted Galactic rotation curve of milky way

When looking at the curve, this is not what we expect to see. Our data is linear where the frequency of appearance of points increases near the end of the curve. As we are only looking at the innermost section, we should have fluctuations in our data.

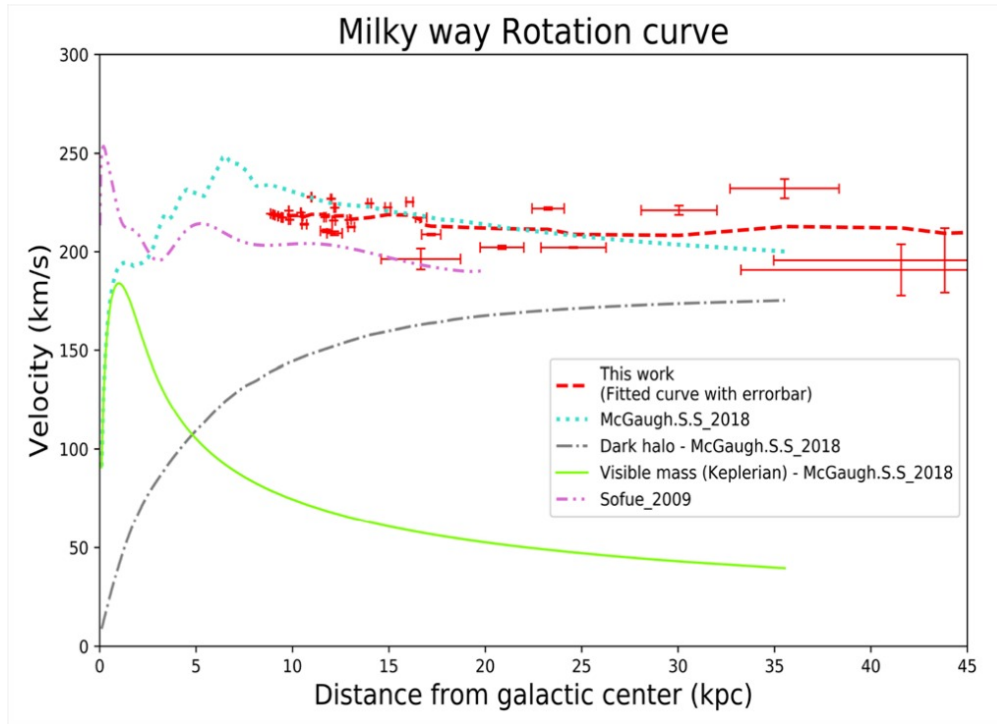


Fig. 8.—: (B Pandian et al 2022)

If we compare our result to figure 8, the milky way rotation curve diagram. we realize that we are plotting

the start of the light green line, which follow into the light blue line.

5. Conclusion and Discussion

In this report, we were able to find and plot the Galactic Rotation Curve by using the data measured from the MP Horn telescope at UofT. We then found the 21cm line and radial velocity associated with various galactic longitudes using the Doppler equation. We finally used the tangent point method to find the rotational velocity and galactocentric radius. Finally we plot these two variables against each other to give us the Galactic Rotation Curve.

As a conclusion to this experiment, we found that in certain cases our lump was not perfect due to having signal from noise or from unknown sources that are affecting our data. Additionally, when comparing our galactic rotation curve to a more detailed version (figure 8), we find that our curve is too linear. However, similarly to the figure 8, the rate of change for the rotational velocity does decrease as the galactocentric radius increases. This nearly perfect linearity is most likely due to the large amounts of reductions that were made to have a data that was focused on the 21 cm only.

There are many points where error and uncertainties could have been created throughout our intake and manipulation of data. Having used a telescope that is in the heart of a large city, signal noise is extremely large causing many issues. We also find that various longitudes will have different data points so need specific darks. The average dark file works for large and consistent peaks but does not take into account specific noise at a random angle. Finally, an additional improvement would be to try a different approach to find the rotational velocity and galactocentric radius, such as the Vector velocity method ([B Pandian et al 2022](#)).

REFERENCES

- B Pandian et al. 2022, galaxy Rotation Curve Measurements with Low Cost 21 cm Radio Telescope, [Accessed: March 8, 2022]
- Sofue, Y., & Rubin, V. 2001, Annual Review of Astronomy and Astrophysics, 39, 137–174

6. Appendix

6.1. Time and Galactic coordinates of measurements

Time (Toronto)	Gal. Lon ($^{\circ}$)
10:35:00 EST	10.00
10:36:30 EST	13.04
10:38:00 EST	16.07
10:39:30 EST	19.11
10:41:00 EST	22.15
10:42:30 EST	25.19
10:44:00 EST	28.22
10:45:30 EST	31.26
10:47:00 EST	34.30
10:48:30 EST	37.33
10:50:00 EST	40.37
10:51:30 EST	43.41
10:53:00 EST	46.44
10:54:30 EST	49.48
10:56:00 EST	52.52
10:57:30 EST	55.56
10:59:00 EST	58.59
11:00:30 EST	61.63
11:02:00 EST	64.67
11:03:30 EST	67.70
11:05:00 EST	70.74
11:06:30 EST	73.78
11:08:00 EST	76.81
11:09:30 EST	79.85
11:11:00 EST	82.89
11:12:30 EST	85.93
11:14:00 EST	88.96

Table 2:: Table of Time and Galactic coordinates of measurments

6.2. Code